Day-11 Oct-13

ROUTING FUNDAMENTALS (PART I)

PART-I

4) Which field of IPv4 header is variable in length?
a) Options b) Header Checksum c) Total Length d) IHL

5) Which bit will be set to 1 on all IPv4 packet fragment except the last fragment?
a) Fragment offset bit b) More fragment bit c) Don't Fragment Bit
d) Packet fragment bit

DAY-11

What is Routing?

When router receives packet it's their job to send it to correct destination. For this they use routing.

- Routing is the process that routers use to determine the path that IP packets should ~~fol~~ take over a network to reach their destination.
- जैसी switch में MAC-Addre रहता Table में store जहाँ रहती है
Routers store route to all their known destination in a routing table
- When routers receive packet, they look in routing table to find best route to forward that packet.

- There are two main routing methods, (methods used by routers to learn rou

Dynamic Routing: Routers use dynamic routing protocols (such as OSPF) to share routing information with each other automatically & build their own routing tables.

Static Routing: A network engineer/admin manually configures routes on router.

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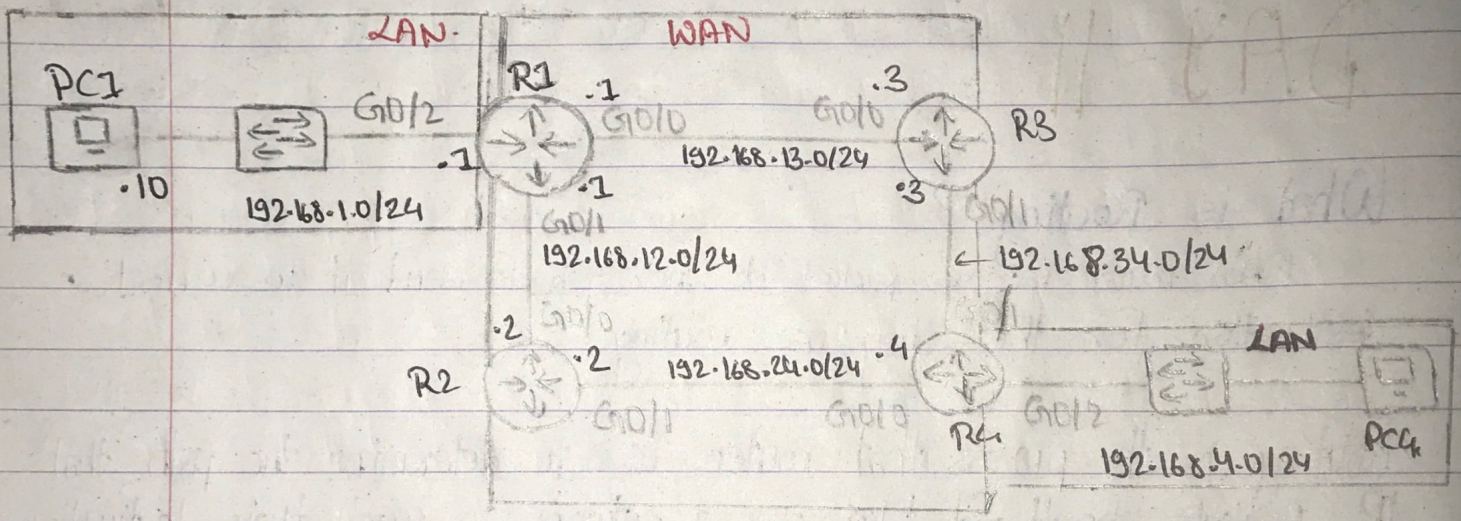
Route is basically instruction to router. It tells router, to send packet to a destination X, you should send the packet to next hop Y.

> next hop = next router on path to destination

→ or if destination is directly connected to router, send the packet directly to destination.

→ or, if the dst. is router's own IP address, receive the packet for yourself don't forward it.

Example Network for this lecture.



Here 4 routers represent WAN. R1 & R4 each have 1 LAN.
In this lecture, we'll focus on two types of routes automatically added to router's routing table (not static or dynamic).

Configuring IP address on R1

- Enter global config mode
- int g0/0 g0/2.
- ip address 192.168.10.1 255.255.255.0
- no shutdown

- To see ip routing table use the following command 'show ip route'

There are two main sections to output.

i) Code at top
Code shows different protocol which routers can used to learn route. as well as codes that represent them in routing table
Eg - L (local) C (Connected)

(ii) Actual routes at bottom
They represent Even if we haven't configured any route on R1, it already has 2 routes.
When we configure IP add on an interface & enable it with 'no shutdown' command 2 routes per interface will automatically be added to the routing table.
A 'connected' route & 'local' route.
These connected & local are neither static nor dynamic

Connected & local routes.

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.1.0/24 is directly connected, Gigabit Ethernet 0/2
L 192.168.1.1/32 is directly connected, Gigabit Ethernet 0/2

Connected Route - A connected route is a route to the network, interface is connected to.

R1 G0/2 IP = 192.168.1.1/24.
Network Address = 192.168.1.0/24 (Distination of cnnctd route).
It provides route to all hosts in network (From .0.1 to .0.255)
R1 knows: if it should send a packet to any host in 192.168.1.0/24 network, it should send it out G0/2.

Local Route: A local route is a route to the exact IP address configured on the interface.

- A /32 netmask is used to specify the exact IP of interface.
  ↳ /32 means all 32 bits are fixed, they can't be changed.

- Even though R1's G0/2 is configured as 192.168.1.1/24 the connected route is 192.168.1.1/32

- R1 knows: If I'll receive a packet destined for this IP addres, the message is for me.

NOTE: A route matches a packet's destination if the packet's destination IP add is part of network, specified on route.

Eg: If R1 receives packet with dst IP = 192.168.2.1 it will drop the packet as no matching route.

192.168.1.1/32 matches only 192.168.1.1.

Route Selection

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.1.0/24 is directly connected, Gigabit Ethernet 0/2
L 192.168.1.1/32 is directly connected, Gigabit Ethernet 0/2

Now,
Suppose R1 received packet with dst IP = 192.168.1.1.

- A packet destined for 192.168.1.1 is matched by both (local & connected) Route. ie by 192.168.1.0/24 & 192.168.1.1/32

So, which route will R1 use for destined 1.1?

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It will choose the most specific matching route i.e 192.168.1.1/32

Most specific matching route = the matching route with longest prefix length

- The route to 192.168.1.0/24 includes 256 different IP address (192.168.1.0 - 192.168.1.255)
- The route to 192.168.1.1/32 includes only one IP add 192.168.1.1
  So, this route is more specific

In A, the top one says 192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks. It means in routing table, there are two routes to subnets that fit within 192.168.1.0/24 Class C network, with two different netmasks (/24 & /32).

Let's see some example

1) R1 receives pckt with dstn IP = 192.168.1.1. Most specific match is 192. local route, so it will receive packet for itself.
2) Packet 2 host dst. IP = 192.168.13.3. So, R1 will send packet to destination which is connected to G0/0 interface
3) Packet 3 is destined for 192.168.4.10. As no matching route R1 will drop this packet.
   Routers never flood packet even if dst is unknown.

QUIZ.

1) The IP add. configured on router interface will appear in routing table as what kind of route?
   a) Static b) Connected c) ✓ Local d) Dynamic.

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2) What will R1 do, if dst IP add = 192.168.3.25? (See video for routing table)

a) It will drop table packet.
b) It will receive packet for itself.
c) It will forward the packet out of G0/0 interface.
d) It will forward packet out of G0/2 interface.

3) Which of the following two behaviours of routers & switches are true?

a) Routers flood packet with unknown dst
b) Switches flood frames with unknown dstn
c) Routers drop packet with unkn. dst
d) Switches drop frames with unkn dstn.

4) Which two types of routes are automatically added to routing table when you configure IP address on interface & enable it.)

a) C,L b) C,S c) L,S d) L,D.

5) Examine R1's routing table, If R1 receives pkt for 10.0.1.23, how many routes match & which is most specific matching route

a) One matching route: 10.0.1.0/24
b) One matching route: 10.0.1.23/32
c) 2 matching route: Most sp: 10.0.1.23/32.
d) 2 matching route, Most sp: 10.0.60/24

NOTE: Remember, if an interface is disabled the connected & local routes for that interface won't appear in routing table. So, don't forget to use "no shutdown" command.

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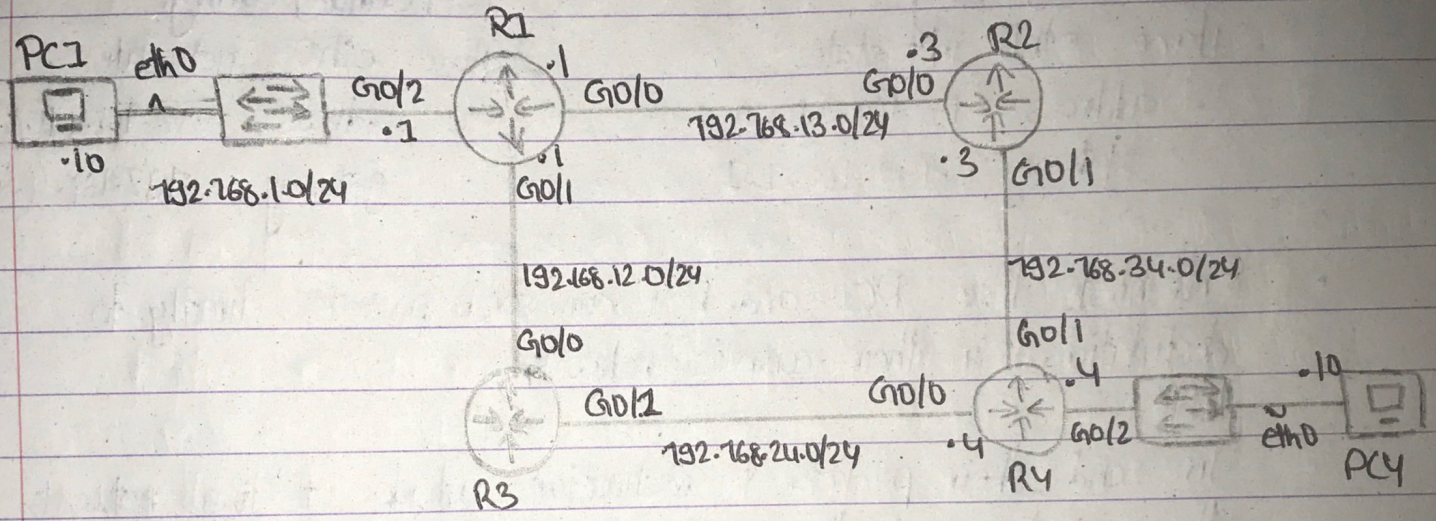
Oct-14

Day-11 (Part-II)

Static Routing

If we want the routers to be able to send traffic to destinations that aren't directly connected to router itself, Local & Connected routes aren't enough.

Static routes enables routers to send packets to remote destinations, that aren't directly connected to router itself.



After configuring all 4 router's IP address, Local & Connected routes for each interface are automatically added in routing table.

Now, let's see R4. It knows how to reach its own IP address & destination in its connected networks, but doesn't know how to reach destination in remote networks. It doesn't know destination on following

192.168.1.0/24
192.168.12.0/24
192.168.13.0/24

Now, if it receives a packet destine for PC4, it has route to PC4, so it would be able to forward packet to PC4

However, if it received a packet destined for PC1, 192.168.1.10, it doesn't have any route that match that destination IP, so it would have to drop the packet.

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Routing Packets: Default Gateway

PC1 & PC4 are linux host & their default interface name is 'eth0' (Ethernet0).

PC1 (Linux) Config
iface eth0 inet static
address: 192.168.1.10/24
gateway: 192.168.1.1

PC4 (Linux) config.
iface eth0 inet static
address 192.168.4.10/24
gateway 192.168.4.4

- End hosts like PC1 and PC4 can send packets directly to destinations in their connected network.
- To send ~~their~~ packets to destinations outside of local networks, they must send the packets to their default gateway

Eg: (To send packet outside of LAN, PC1 will send packet to R1)

- The default gateway is also called default route.
  - It is a route to 0.0.0.0 /0 = all netmask bits sets to 0. Includes all addresses from 0.0.0.0 to 255.255.255.25
  - It's a route that includes everything.
- Default route is the 'least specific' route possible because it includes all IP ~~fic~~ addresses.
- 0.0.0.0/0 includes over 4 billion IP addresses.
- Defaut route says "if ~~there~~ there aren't any more specific matches for this packet, don't drop it, send it via this route instead

End hosts usually have no need for any more specific routes, they just need to know to send packet out of LAN, I should send them to my default gateway.

If PC1 wants to send packet to PC4 the ~~source IP header~~ following would be source & destination

Src IP:	192.168.1.10
Dst IP:	192.168.4.10
Dst MAC:	R1 G0/2 MAC
Src MAC:	PC1 eth0 MAC.

Dst MAC चाहिए default gateway को हुएइ coz d. PC4 out of LAN छ R1.
To know R1's G0/2 mac, PC1 ले ARP Req send गर्छ हुन्छ

Static Routes

When, R1 receive packet, it de-encapsulates it & look inside the packet
- It will the check routing table for most-specific matching route, It has to make decision about what it will do with this packet
- (अहिले cannot 2 local Routes भए पनि, so matching route for PC4 हुँदैन)
- It will drop packet.

To forward it, R1 needs a route to dst. IP network, 192.168.4.0/24
-Routes are instructions: To send a packet to destination in network 192.168.40/24, forward the packet to next hop Y.

- There are two possible paths, ~~path~~ packets from PC1 to PC4 can take
1) PC1 → R1 → R3 → R4 → PC4
2) PC1 → R1 → R2 → R4 → PC4.

- Let's see examples using R3.

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If there are multiple paths to dstn; we can choose which one the routers should use.
- It is possible to configure routers to use these multiples paths differently.
- Instead of choosing one path they can load-balance between 1 & 2, or they could use path1 as main path & path2 as back-up.

Static Routes configurate

- For PC1 to communicate with PC4 & vice-versa, its just not just R1 that need another route.
- Each router in the path needs two routes, a route to 192.168.1.0/24 & a route to 192.168.4.0/24
  ↳ This ensures two-way reachability (PC1 can send packets to PC4, PC4 can send packets to PC1).
- Routers don't need routes to all networks in path to destination.
  Eg. R1 doesn't need route to 192.168.34.0/24. It just need to know to send packet to PC4, It shall send packet to R3.
- R1 already has connected routes to 192.168.1.0/24. R4 already has connected route to 192.168.4.0/24.
  ↳ The other routes must be manually configured. (using Static routes)
- The planning of routes we'll configure is as follow:

- To allow PC1 & PC4 to communicate with each other, let's configure these static routes on R1, R3 & R4

Router	Destination	Next-Hop
R1	192.168.1.0/24	Connected
	192.168.4.0/24	192.168.13.3
R3	192.168.1.0/24	192.168.13.1
	192.168.4.0/24	192.168.34.4
R4	192.168.1.0/24	192.168.34.3
	192.168.4.0/24	Connected

Heti Commands To Configure

- Enter global-config mode.
  'ip route ip-address netmask next-hop'
  R1(config)# ip route 192.168.4.0 255.255.255.0 192.168.13.3

- To check routing table.
  do 'do show ip route'
  It will show static route with 'S' as code.
  S 192.168.4.0/24 [1/0] via 192.168.13.3
  In sq. bracket there is [1/0]. They are routes "administrative distance" & "metric".

- There's also another method to configure static route.
  ip route ip-address netmask exit-interface
  ip route ip-address netmask ex-interface next-hop

When we specify only exit-interface on static int-route, it says destination network is directly connected
Static-route on which we specify exit-interface only rely on feature called 'proxy-arp)

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Gateway of last resort is another name for default gateway.

Travelling of pkt from PC1 to PC4.

PC1 → R1

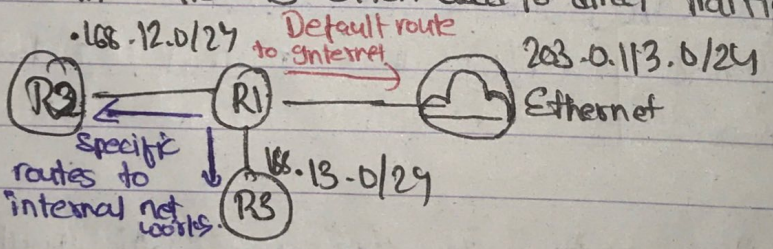
Src: IP : 168.1.10
Dst IP: 168.4.10
Src MAC: PC1 eth0 MAC
Dst MAC: R1 G0/2 MAC.

	Src IP	Dst IP	Src MAC	Dst MAC
R1 → R3	192.168.1.10	168.4.10	R1 G0/0	R3 G0/0
R3 → R4	168.1.10	168.4.10	R3 G0/1	R4 G0/1

The SRC & DST IP address, doesn't change throughout process

Default Route

- A default route is a route to 0.0.0.0/0 & is least specific route
- A default route is often used to direct traffic to internet.



- More specific routes are used for destinations in internal corporate network.
- show ip route.
- Gateway of last resort not set (default gateway configure नहीं है)
- (Traffic to destn, outside of internal network is sent to internet)
- Con' To configure default gateway use these commands

'ip route 0.0.0.0 0.0.0.0 203.0.113.2

In ip table its code will be S*. It means this route is candidate to become router's default ~~gateway~~ route.
It's possible to have multiple candidates.

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QUIZ

Q1- Which of following command configures default route on CISCO routers?

a) ip route 0.0.0.0 0.0.0.0 10.1.1.255
b) ip route 0.0.0.0/0 10.1.1.254
c) ip route 0.0.0.0 255.255.255.255 10.1.1.255
d) ip route 0.0.0.0/32 10.1.1.254

Q2) Examine R1's routing table. Which int. will be used to forward packets destined for 8.8.8.8?

a) G 0/0 b) G 0/1 c) G0/2 d) It will drop the packet.

Q4) Examine the following static route in R1's table. What command was used to configure this route.

172.20.0.0/16 is directly connct, g0/1.

a) ip route 172.20.0.0 255.255.0.0 g0/1
b) " " " 255.255.0.0
c) " " " g0/1 172.20.0.1
d) " " " 255.255.0.0 g0/1.

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